

A SELF-CONTAINED RESEARCH NOTE

Apéry Arrays and Conjectural Zeta Accelerations

Proofs for every shifted diagonal in OEIS A143007,
and for both conjectural diagonal families in A108625

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Abstract

The OEIS array A143007 contains a conjectural one-parameter family of rapidly convergent series for $\zeta(3)$. Its main diagonal consists of the classical Apéry numbers. We prove the entire family, not merely a finite collection of numerical instances. The proof constructs a rational potential on the nonnegative integer lattice: two explicitly certified contiguous identities make its edge increments independent of the chosen path, and elementary estimates identify its limit as $\zeta(3)$. The same method proves both conjectural diagonal families for $\zeta(2)$ in A108625, at every upper and lower offset, which is the full range for which the entry states them. We also obtain exact finite identities, signed rational error bounds, all monotone-path expansions, the classical Apéry recurrences, a three-term recurrence along *every* shifted diagonal, certified decimal prefixes, and sharp fixed-offset asymptotics. The accompanying code checks 34,910 identities in exact arithmetic, in a union suite with a second independent implementation, and verifies the finite symbolic certificates. The analytic proof does not depend on those tests.

Status of the claim. The targeted formulas are still described conjecturally in the OEIS versions consulted on the date above. This note supplies a complete proof of those statements. It does not claim that their underlying Apéry acceleration mechanism, or every consequence developed here, is new. In particular, David’s conservative-matrix-field paper already develops closely related lattice constructions for these constants and contains, up to its duality and sign conventions, the conjugate polynomials occurring below [5, Examples 16, 19, 25; §5]; see [section 1](#).

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1 The problem and its mathematical setting

Apéry’s sequences are an especially appropriate setting for an OEIS conjecture: they connect explicit integer sums, second-order recurrences, rational approximation, and special values of the Riemann zeta function. The familiar sequence associated with $\zeta(3)$ is

$$A_n = \sum_{j=0}^n \binom{n}{j}^2 \binom{n+j}{j}^2 = 1, 5, 73, 1445, 33001, 819005, \dots, \quad (1.1)$$

where the displayed list begins at $n = 0$. This is [A005259](#). Its $\zeta(2)$ companion, [A005258](#), is

$$b_n = \sum_{j=0}^n \binom{n}{j}^2 \binom{n+j}{j} = 1, 3, 19, 147, 1251, 11253, \dots \quad (1.2)$$

These are not incidental integer sequences: they are the denominators in the classical Apéry approximation constructions [6, 3, 4, 7, 8].

The two-variable arrays studied below have these sequences as their main diagonals. They also have an independent geometric interpretation. Let

$$A_m = \{(x_1, \dots, x_{m+1}) \in \mathbb{Z}^{m+1} : \sum_i x_i = 0\}, \quad \|x\| = \frac{1}{2} \sum_i |x_i|.$$

The array A108625 counts balls in A_m ; A143007 counts balls in $A_m \times A_m$ for the sum of the two norms [2, 1, 9]. Our proof will use finite binomial sums rather than assume any geometric facts about these lattices.

1.1 Exactly which conjectures are proved?

The formula section of A143007 explicitly labels its shifted-diagonal $\zeta(3)$ formula a “Conjectural result for other diagonals” [1]. We prove it for every integer offset $k \geq 0$ in [theorem 2.1](#). The entry attributes its construction to Peter Bala and dates the original sequence to July 22, 2008.

The formula section of A108625 likewise carries two separate labels, “Conjectural result for superdiagonals” and “Conjectural result for subdiagonals”, each stated for $k = 0, 1, 2, \dots$ and each attributed to Peter Bala [2]. These are exactly [theorems 2.2](#) and [2.3](#). Both families are therefore already *stated* in the entry for every nonnegative offset; what is supplied here is their proof, including the necessary distinction between the upper and lower diagonals. The entry’s earlier comments of July 18, 2008 describe the main diagonal and its two immediate neighbors, which are the $k = 0$ and $k = 1$ cases.

For reproducibility, the consulted internal revisions are A143007, revision 75, and A108625, revision 98; both carry a May 30, 2026 revision timestamp. A conjectural label in a database is evidence about the entry’s wording, not a guarantee that no proof exists elsewhere.

That caveat is not hypothetical here. David’s paper on the conservative matrix field develops related lattice constructions for these constants and contains, up to its duality and sign conventions, the same conjugate polynomials that appear below; its Section 5 establishes convergence and relates a diagonal trajectory to Apéry’s irrationality argument [5, Examples 16, 19, 25; §5]. This substantial overlap precludes any unsupported claim that the method, its polynomials, or all of its consequences originate in this note. An OEIS label can persist after an identity follows from a broader theorem in the literature. Two-dimensional and staircase versions of Apéry acceleration are classical in

addition; Cohen's account explicitly develops that viewpoint [7, §§3–4]. The contribution of this note is a direct, self-contained proof of the exact OEIS statements at the level of finite binomial sums, together with explicit finite certificates and error analysis.

2 The arrays, their indexing, and the main results

All indices in this note are *square-array* indices. They are not indices in the one-dimensional antidiagonal flattening used on the OEIS pages. Set

$$L(m, j) = \binom{m}{j} \binom{m+j}{j}, \quad L(m, j) = 0 \quad \text{if } j > m, \quad (2.1)$$

and, for $m, n \geq 0$, define

$$T_{m,n} = \sum_{j=0}^{\min(m,n)} L(m, j) L(n, j), \quad (2.2)$$

$$C_{m,n} = \sum_{j=0}^{\min(m,n)} L(m, j) \binom{n}{j}. \quad (2.3)$$

The array T is A143007, and C is A108625, with the dimension parameter in the first index and the radius in the second. Both OEIS entries call their own array T ; we reserve T for the symmetric array and write C for A108625, so that the two can appear in one formula. The two rational potentials built from them below are written Φ and Ψ , never U or V , because U is a natural name for the A108625 array itself. Indeed,

$$\binom{m+j}{2j} \binom{2j}{j} = L(m, j),$$

so (2.2) immediately agrees with the OEIS formula. The equivalence of (2.3) with the entry's binomial formula and row generating function is proved in [section B](#).

The initial portions are

$T_{m,n}$	$n = 0$	1	2	3	4	$C_{m,n}$	$n = 0$	1	2	3	4
$m = 0$	1	1	1	1	1	$m = 0$	1	1	1	1	1
1	1	5	13	25	41	1	1	3	5	7	9
2	1	13	73	253	661	2	1	7	19	37	61
3	1	25	253	1445	5741	3	1	13	55	147	309

Thus $T_{m,n} = T_{n,m}$, but generally $C_{m,n} \neq C_{n,m}$. In particular, transposing C silently would interchange the two $\zeta(2)$ conjectures and alter their constants and signs.

Both arrays are positive integers and are nondecreasing in each index, as follows directly from their nonnegative summands. Useful boundary values are

$$T_{0,n} = T_{m,0} = C_{0,n} = C_{m,0} = 1, \quad (2.4)$$

$$T_{m,1} = 1 + 2m(m+1), \quad C_{m,1} = 1 + m(m+1), \quad C_{1,n} = 1 + 2n. \quad (2.5)$$

Their diagonals are precisely $T_{n,n} = A_n$ and $C_{n,n} = b_n$.

Write

$$H_k^{(s)} = \sum_{j=1}^k \frac{1}{j^s}, \quad E_k = 2 \sum_{j=1}^k \frac{(-1)^{j+1}}{j^2}, \quad H_0^{(s)} = E_0 = 0.$$

Theorem 2.1 (All shifted $\zeta(3)$ diagonals). *For every integer $k \geq 0$,*

$$\zeta(3) = H_k^{(3)} + \sum_{n=1}^{\infty} \frac{(2n+k)(3n^2+3nk+k^2)}{n^3(n+k)^3 T_{n-1,n+k-1} T_{n,n+k}}. \quad (2.6)$$

The series is positive and converges geometrically. Its partial sum through $n = N$ is strictly below $\zeta(3)$.

At $k = 0$, (2.6) is the classical Apéry series

$$\zeta(3) = 6 \sum_{n=1}^{\infty} \frac{1}{n^3 A_{n-1} A_n}. \quad (2.7)$$

For example, the first genuinely shifted case is

$$\zeta(3) = 1 + \frac{21}{8 \cdot 13} + \frac{95}{216 \cdot 13 \cdot 253} + \cdots.$$

Theorem 2.2 (All upper $\zeta(2)$ diagonals). *For every integer $k \geq 0$,*

$$\zeta(2) = H_k^{(2)} + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(n+k)^2 + (2n+k)^2}{n^2(n+k)^2 C_{n-1,n+k-1} C_{n,n+k}}. \quad (2.8)$$

The series is absolutely convergent. If $S_{N,k}^{(2,+)}$ denotes its partial sum including $H_k^{(2)}$, then $\operatorname{sgn}(\zeta(2) - S_{N,k}^{(2,+)}) = (-1)^N$.

The numerator in (2.8) expands to $5n^2 + 6kn + 2k^2$, which is how the entry writes it.

Theorem 2.3 (All lower $\zeta(2)$ diagonals). *For every integer $k \geq 0$,*

$$\zeta(2) = E_k + \sum_{n=1}^{\infty} (-1)^{n+k+1} \frac{n^2 + (2n+k)^2}{n^2(n+k)^2 C_{n+k-1,n-1} C_{n+k,n}}. \quad (2.9)$$

The series is absolutely convergent. If $S_{N,k}^{(2,-)}$ denotes its partial sum including E_k , then $\operatorname{sgn}(\zeta(2) - S_{N,k}^{(2,-)}) = (-1)^{N+k}$.

Here the numerator expands to $5n^2 + 4kn + k^2$, and the entry writes the overall sign as an explicit factor $(-1)^k$ multiplying a series with sign $(-1)^{n+1}$; this is the same statement.

At $k = 0$ both theorems give

$$\zeta(2) = 5 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2 b_{n-1} b_n}. \quad (2.10)$$

Their $k = 1$ cases are the neighboring-diagonal instances singled out in the entry's comments:

$$\zeta(2) = 1 + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(n+1)^2 + (2n+1)^2}{n^2(n+1)^2 C_{n-1,n} C_{n,n+1}}, \quad (2.11)$$

$$\zeta(2) = 2 - \sum_{n=1}^{\infty} (-1)^{n+1} \frac{n^2 + (2n+1)^2}{n^2(n+1)^2 C_{n,n-1} C_{n+1,n}}. \quad (2.12)$$

For comparison with the entry, these begin

$$1 + \frac{13}{4 \cdot 5} - \frac{34}{36 \cdot 5 \cdot 37} + \cdots, \quad 2 - \frac{10}{4 \cdot 7} + \frac{29}{36 \cdot 7 \cdot 55} - \cdots.$$

3 Finite contiguous identities: the algebraic certificates

The proof is driven by identities involving the four corners of one unit square. There is no experimental recurrence guessing in what follows: each identity is certified by an explicit telescoping summand.

3.1 The $\zeta(3)$ square

Fix integers $m, n \geq 1$ and abbreviate

$$a = T_{m-1,n-1}, \quad b = T_{m,n-1}, \quad c = T_{m-1,n}, \quad d = T_{m,n}, \quad s = (m+n)(m^2 + mn + n^2).$$

Lemma 3.1. *The four corners satisfy*

$$m^3d + n^3a = sc, \tag{3.1}$$

$$n^3d + m^3a = sb. \tag{3.2}$$

Proof. Let $M = \min(m, n)$ and $f_j = L(m, j)L(n, j)$ for $0 \leq j \leq M$. The ratios needed to lower the parameters are

$$\frac{L(m-1, j)}{L(m, j)} = \frac{m-j}{m+j}, \quad \frac{L(n-1, j)}{L(n, j)} = \frac{n-j}{n+j}.$$

They remain valid at an endpoint where a numerator vanishes. Define

$$R_j = \frac{j^4}{(m+j)(n+j)} f_j \quad (0 \leq j \leq M), \quad R_{M+1} = 0.$$

The ratio of consecutive summands gives, for $0 \leq j \leq M$,

$$R_{j+1} = (m-j)(n-j)f_j. \tag{3.3}$$

For $j = M$ both sides vanish; for $j < M$ this follows from

$$\frac{f_{j+1}}{f_j} = \frac{(m-j)(m+j+1)(n-j)(n+j+1)}{(j+1)^4}.$$

The following rational identity is the certificate:

$$\begin{aligned} m^3 + n^3 \frac{(m-j)(n-j)}{(m+j)(n+j)} - s \frac{m-j}{m+j} \\ = 2(m+n) \left\{ \frac{j^4}{(m+j)(n+j)} - (m-j)(n-j) \right\}. \end{aligned} \tag{3.4}$$

It is checked simply by multiplying by $(m+j)(n+j)$ and expanding. Multiplication by f_j turns its right side into $2(m+n)(R_j - R_{j+1})$. Summing from 0 to M proves (3.1), since $R_0 = R_{M+1} = 0$. Interchanging m and n and using the symmetry of T proves (3.2). $\square$

It is worth recording that the single certificate (3.4) can be split into two pieces, one of which needs no telescoper at all. Writing $\rho_m = (m-j)/(m+j)$ and $\rho_n = (n-j)/(n+j)$ for the two lowering ratios, the purely rational identity

$$(m+n)(\rho_n - \rho_m) = (m-n)(\rho_m \rho_n - 1)$$

is termwise zero, and summing it against f_j gives the *antisymmetric* relation

$$(m+n)(b-c) = (m-n)(a-d). \quad (3.5)$$

The remaining, *symmetric* content

$$(m^2 - mn + n^2)(a+d) = (m^2 + mn + n^2)(b+c) \quad (3.6)$$

is certified by the same kernel: multiplying (3.4)'s telescoping bracket by 4 instead of $2(m+n)$ gives exactly the summand identity behind (3.6). Adding $(m+n)$ times (3.6) to $(m^2 + mn + n^2)$ times (3.5) returns (3.1), and subtracting returns (3.2). This is a repackaging of one certificate in different coordinates, not a second proof; we keep [theorem 3.1](#) as the primary form because the potential argument uses it directly.

The conjugate polynomial

$$\bar{s} = (m-n)(m^2 - mn + n^2), \quad s\bar{s} = m^6 - n^6, \quad (3.7)$$

turns [theorem 3.1](#) into the equivalent pair

$$m^3c - n^3b = \bar{s}d, \quad m^3b - n^3c = \bar{s}a, \quad (3.8)$$

obtained by eliminating a or d between (3.1) and (3.2) and using $s\bar{s} = m^6 - n^6$. We use (3.8) in [section 7](#).

3.2 The $\zeta(2)$ square

Now use the same local letters for the corners of C :

$$a = C_{m-1,n-1}, \quad b = C_{m,n-1}, \quad c = C_{m-1,n}, \quad d = C_{m,n}, \quad q = n^2 + (m+n)^2.$$

Lemma 3.2. *The four corners satisfy*

$$2n^2d - m^2a = qb, \quad (3.9)$$

$$m^2d + 2n^2a = qc. \quad (3.10)$$

Proof. Put $g_j = L(m, j) \binom{n}{j}$, $M = \min(m, n)$, and set

$$r_m = \frac{m-j}{m+j}, \quad r_n = \frac{n-j}{n}, \quad R_j = \frac{j^3}{m+j} g_j \quad (0 \leq j \leq M), \quad R_{M+1} = 0.$$

Here r_m and r_n are the ratios obtained by lowering m and n , respectively. As in the preceding proof,

$$R_{j+1} = (m-j)(n-j)g_j,$$

because

$$\frac{g_{j+1}}{g_j} = \frac{(m-j)(m+j+1)(n-j)}{(j+1)^3} \quad (j < M).$$

Two elementary rational identities are

$$m^2 + 2n^2r_m r_n - qr_m = 2 \left\{ \frac{j^3}{m+j} - (m-j)(n-j) \right\}, \quad (3.11)$$

$$2n^2 - m^2r_m r_n - qr_n = \frac{2(m+n)}{n} \left\{ \frac{j^3}{m+j} - (m-j)(n-j) \right\}. \quad (3.12)$$

Multiply each identity by g_j , sum, and use $R_0 = R_{M+1} = 0$. The resulting identities are (3.10) and (3.9). All denominators used here are nonzero because $m, n \geq 1$. $\square$

The quadratic square has the same conjugate structure. With

$$\bar{q} = m^2 - 2mn + 2n^2, \quad q\bar{q} = m^4 + 4n^4, \quad (3.13)$$

eliminating a or d between (3.9) and (3.10) gives

$$2n^2c - m^2b = \bar{q}a, \quad m^2c + 2n^2b = \bar{q}d. \quad (3.14)$$

Multiplying the first by d , the second by a , and equating yields the square condition for the signed weights of section 5 in one line. We verify that condition directly instead, in (5.2); (3.14) is recorded because it is the shortest statement of what the two corner identities say jointly.

These three certificates are sufficient for the whole infinite family of conjectures. Their role is structural: a local identity will allow an infinite series to be evaluated by changing its lattice path.

4 A rational potential proving the $\zeta(3)$ conjecture

4.1 Path independence on the integer lattice

A horizontal edge into (m, n) is the step $(m-1, n) \rightarrow (m, n)$, and a vertical edge into (m, n) is $(m, n-1) \rightarrow (m, n)$. If their weights are $h(m, n)$ and $v(m, n)$, respectively, the square condition is

$$h(m, n-1) + v(m, n) = v(m-1, n) + h(m, n). \quad (4.1)$$

It says that the two routes from $(m-1, n-1)$ to (m, n) have the same weight.

Lemma 4.1 (Finite path independence). *If (4.1) holds for every $m, n \geq 1$, then every monotone lattice path from $(0, 0)$ to (m, n) has the same sum of edge weights. Consequently there is a potential $F_{m,n}$ with $F_{0,0} = 0$ and*

$$F_{m,n} - F_{m-1,n} = h(m, n), \quad F_{m,n} - F_{m,n-1} = v(m, n).$$

Proof. Encode a monotone path as a word with m horizontal steps and n vertical steps. Adjacent horizontal-vertical pairs can be interchanged without changing the sum, by (4.1). Such interchanges connect any two words with the same numbers of each step. Define $F_{m,n}$ as this common sum. Appending a last edge proves the asserted increments. $\square$

For T , assign the positive rational edge weights

$$h_3(m, n) = \frac{1}{m^3 T_{m-1,n} T_{m,n}}, \quad v_3(m, n) = \frac{1}{n^3 T_{m,n-1} T_{m,n}}. \quad (4.2)$$

Each expression is used only when the corresponding edge exists. With the local corner notation of theorem 3.1, condition (4.1), after clearing positive denominators, is

$$c(n^3 d + m^3 a) = b(m^3 d + n^3 a).$$

Both sides are $sb c$, by theorem 3.1. Hence the weights are path independent.

Let their potential be $\Phi_{m,n}$. Taking the two boundary routes yields the explicit finite identity

$$\Phi_{m,n} = H_n^{(3)} + \sum_{i=1}^m \frac{1}{i^3 T_{i-1,n} T_{i,n}} = H_m^{(3)} + \sum_{j=1}^n \frac{1}{j^3 T_{m,j-1} T_{m,j}}. \quad (4.3)$$

In particular $\Phi_{m,n}$ is rational, symmetric, and strictly increasing when either coordinate increases by one.

4.2 Identifying the limit without assuming an Apéry theorem

Lemma 4.2. *For all $m, n \geq 0$, $\Phi_{m,n} < \zeta(3)$. Moreover, $\Phi_{m,n} \rightarrow \zeta(3)$ whenever $\max(m, n) \rightarrow \infty$.*

Proof. The second expression in (4.3) and monotonicity of T imply

$$0 \leq \Phi_{m,n} - H_m^{(3)} \leq \frac{\zeta(3)}{T_{m,1}} = \frac{\zeta(3)}{1 + 2m(m+1)}.$$

This tends to zero as $m \rightarrow \infty$, uniformly in n ; thus for each fixed n , the strictly increasing sequence $\Phi_{m,n}$ tends to $\zeta(3)$. In particular every finite value is strictly below $\zeta(3)$. The two expressions in (4.3) give

$$H_{\max(m,n)}^{(3)} \leq \Phi_{m,n} < \zeta(3).$$

The last assertion now follows by squeezing. $\square$

It follows, incidentally, that *every* infinite monotone path starting at $(0, 0)$ yields a positive series with sum $\zeta(3)$. Rows and diagonals are particular paths, not separate approximation phenomena.

4.3 The diagonal increment

The decisive cancellation is finite. A horizontal step followed by a vertical step, together with (3.2), gives

$$\begin{aligned} \Phi_{m,n} - \Phi_{m-1,n-1} &= \frac{1}{m^3 ab} + \frac{1}{n^3 bd} \\ &= \frac{n^3 d + m^3 a}{m^3 n^3 abd} \\ &= \frac{(m+n)(m^2 + mn + n^2)}{m^3 n^3 T_{m-1,n-1} T_{m,n}}. \end{aligned} \tag{4.4}$$

The intermediate corner b cancels because the contiguous identity makes its numerator a multiple of b . This explains the apparently special cubic polynomial in the conjectured summand.

Proof of theorem 2.1. Apply (4.4) at the points $(n, n+k)$ and sum the resulting identity for $n = 1, \dots, N$. Since $\Phi_{0,k} = H_k^{(3)}$, the exact result is

$$\Phi_{N,N+k} = H_k^{(3)} + \sum_{n=1}^N \frac{(2n+k)(3n^2 + 3nk + k^2)}{n^3(n+k)^3 T_{n-1,n+k-1} T_{n,n+k}}. \tag{4.5}$$

Let $N \rightarrow \infty$ and apply theorem 4.2. All terms are positive, and the partial sums are below the limit. Geometric convergence follows from the explicit bounds in section 6, independently of the sharper asymptotics in section 8. $\square$

5 A signed rational potential for the $\zeta(2)$ conjectures

The array C is not symmetric, and the correct weights necessarily remember the first coordinate:

$$h_2(m, n) = \frac{2(-1)^{m+1}}{m^2 C_{m-1,n} C_{m,n}}, \quad v_2(m, n) = \frac{(-1)^m}{n^2 C_{m,n-1} C_{m,n}}. \tag{5.1}$$

The square condition (4.1), after clearing denominators and a common sign, becomes

$$c(2n^2d - m^2a) = b(m^2d + 2n^2a). \quad (5.2)$$

By theorem 3.2, both sides are qbc . Hence these weights define a rational potential $\Psi_{m,n}$. The boundary values now differ on the two axes:

$$\Psi_{0,n} = H_n^{(2)}, \quad \Psi_{m,0} = E_m.$$

The two boundary routes give

$$\Psi_{m,n} = H_n^{(2)} + 2 \sum_{i=1}^m \frac{(-1)^{i+1}}{i^2 C_{i-1,n} C_{i,n}} = E_m + (-1)^m \sum_{j=1}^n \frac{1}{j^2 C_{m,j-1} C_{m,j}}. \quad (5.3)$$

Lemma 5.1. *The potential $\Psi_{m,n}$ tends to $\zeta(2)$ whenever $\max(m, n) \rightarrow \infty$.*

Proof. First, $E_m \rightarrow \zeta(2)$: separating the even terms in the absolutely convergent series gives

$$\sum_{j \geq 1} \frac{(-1)^{j+1}}{j^2} = \zeta(2) - 2 \sum_{j \geq 1} \frac{1}{(2j)^2} = \frac{1}{2} \zeta(2).$$

The second expression in (5.3) gives the uniform estimate

$$|\Psi_{m,n} - E_m| \leq \frac{\zeta(2)}{C_{m,1}} = \frac{\zeta(2)}{1 + m(m+1)}. \quad (5.4)$$

On the other hand, the first expression gives

$$|\Psi_{m,n} - H_n^{(2)}| \leq \frac{2\zeta(2)}{C_{1,n}} = \frac{2\zeta(2)}{2n+1}. \quad (5.5)$$

For (5.5), use $C_{i,n} \geq C_{1,n}$ and $C_{i-1,n} \geq 1$ for every $i \geq 1$. Either estimate proves the limit when its corresponding coordinate tends to infinity, uniformly in the other. Together they prove the assertion for $\max(m, n) \rightarrow \infty$. $\square$

As before, every infinite monotone path produces a convergent expansion for the same constant, although the edge weights now have signs. To combine the two edges of a square, use (3.9):

$$\begin{aligned} \Psi_{m,n} - \Psi_{m-1,n-1} &= \frac{2(-1)^{m+1}}{m^2ab} + \frac{(-1)^m}{n^2bd} \\ &= (-1)^{m+1} \frac{2n^2d - m^2a}{m^2n^2abd} \\ &= (-1)^{m+1} \frac{n^2 + (m+n)^2}{m^2n^2C_{m-1,n-1}C_{m,n}}. \end{aligned} \quad (5.6)$$

Proof of theorems 2.2 and 2.3. Sum (5.6) along the points $(n, n+k)$, starting from $(0, k)$. This gives the exact finite identity

$$\Psi_{N,N+k} = H_k^{(2)} + \sum_{n=1}^N (-1)^{n+1} \frac{(n+k)^2 + (2n+k)^2}{n^2(n+k)^2 C_{n-1,n+k-1} C_{n,n+k}}. \quad (5.7)$$

Instead sum along $(n + k, n)$, starting from $(k, 0)$. The result is

$$\Psi_{N+k,N} = E_k + \sum_{n=1}^N (-1)^{n+k+1} \frac{n^2 + (2n + k)^2}{n^2(n + k)^2 C_{n+k-1,n-1} C_{n+k,n}}. \quad (5.8)$$

Take $N \rightarrow \infty$ and apply [theorem 5.1](#). Absolute convergence already follows by bounding the positive arrays below by 1: for fixed k , the absolute summands are $O(n^{-2})$. The exact signs of the errors follow from (6.5) below. $\square$

The additive constant E_k in the lower formula is not interchangeable with $H_k^{(2)}$. Likewise its extra sign $(-1)^k$ is not optional. Both are forced by the boundary values and the orientation of the potential.

6 Exact remainders and certified rational intervals

A proof of convergence is stronger when it also gives computable error bounds. Here the potentials make the error an explicit tail in either coordinate direction, so no numerical evaluation of a zeta function is required for certification.

6.1 The positive $\zeta(3)$ error

For fixed n , extend the horizontal path from (m, n) to infinity. By [theorem 4.2](#),

$$\zeta(3) - \Phi_{m,n} = \sum_{i=m+1}^{\infty} \frac{1}{i^3 T_{i-1,n} T_{i,n}} = \sum_{j=n+1}^{\infty} \frac{1}{j^3 T_{m,j-1} T_{m,j}}. \quad (6.1)$$

Both representations are positive. For $m \geq 1$, monotonicity gives

$$\begin{aligned} \frac{1}{(m+1)^3 T_{m,n} T_{m+1,n}} &\leq \zeta(3) - \Phi_{m,n} \\ &\leq \frac{\zeta(3) - H_m^{(3)}}{T_{m,n} T_{m+1,n}} < \frac{1}{2m^2 T_{m,n} T_{m+1,n}}. \end{aligned} \quad (6.2)$$

The last inequality uses $\sum_{i>m} i^{-3} < \int_m^{\infty} t^{-3} dt = 1/(2m^2)$. There is a second bound with the coordinates interchanged. A third form uses the tail whose lower index is the *larger* coordinate and then replaces both array factors by $T_{m,n}$: for $M = \max(m, n) \geq 1$,

$$0 < \zeta(3) - \Phi_{m,n} < \frac{1}{2M^2 T_{m,n}^2}. \quad (6.3)$$

Thus, for $m, n \geq 1$, the explicitly rational number

$$B_3(m, n) = \min \left\{ \frac{1}{2m^2 T_{m,n} T_{m+1,n}}, \frac{1}{2n^2 T_{m,n} T_{m,n+1}}, \frac{1}{2 \max(m, n)^2 T_{m,n}^2} \right\} \quad (6.4)$$

satisfies

$$\Phi_{m,n} < \zeta(3) < \Phi_{m,n} + B_3(m, n).$$

All three expressions are proved bounds, so the minimum is legitimate. Monotonicity of T in fact makes the branch of (6.4) belonging to the larger coordinate at least as sharp as (6.3): along a fixed diagonal the gain is the factor $T_{N,N+k+1}/T_{N,N+k} \rightarrow q_3 = 3 + 2\sqrt{2} = 5.828\dots$, by [theorem 8.1](#), which is worth about 0.77 of a decimal digit. For the conjectural series, substitute $(m, n) = (N, N + k)$.

6.2 The signed $\zeta(2)$ error

Taking a vertical tail in (5.3) gives the exact identity

$$\zeta(2) - \Psi_{m,n} = (-1)^m \sum_{j=n+1}^{\infty} \frac{1}{j^2 C_{m,j-1} C_{m,j}}. \quad (6.5)$$

Every summand on the right after removal of $(-1)^m$ is strictly positive. Hence the sign is exactly $(-1)^m$, not merely eventually so. For $n \geq 1$, the vertical estimates are

$$\frac{1}{(n+1)^2 C_{m,n} C_{m,n+1}} \leq |\zeta(2) - \Psi_{m,n}| < \frac{1}{n C_{m,n} C_{m,n+1}}. \quad (6.6)$$

The upper bound follows from $\sum_{j>n} j^{-2} < 1/n$.

There is a complementary horizontal tail:

$$\zeta(2) - \Psi_{m,n} = 2 \sum_{i=m+1}^{\infty} \frac{(-1)^{i+1}}{i^2 C_{i-1,n} C_{i,n}}. \quad (6.7)$$

Its magnitudes strictly decrease to zero, since i^2 strictly increases and both array factors are nondecreasing. The alternating-series bound therefore gives

$$|\zeta(2) - \Psi_{m,n}| \leq \frac{2}{(m+1)^2 C_{m,n} C_{m+1,n}}. \quad (6.8)$$

Replacing $C_{m,n+1}$ by $C_{m,n}$ in (6.6) gives the weaker but equally valid form $|\zeta(2) - \Psi_{m,n}| < 1/(nC_{m,n}^2)$. For $n \geq 1$, define

$$B_2(m, n) = \min \left\{ \frac{2}{(m+1)^2 C_{m,n} C_{m+1,n}}, \frac{1}{n C_{m,n} C_{m,n+1}}, \frac{1}{n C_{m,n}^2} \right\}. \quad (6.9)$$

Then $\Psi_{m,n}$ and $\Psi_{m,n} + (-1)^m B_2(m, n)$ are endpoints of a closed rational interval containing $\zeta(2)$. The parity gives the correct ordering of its endpoints. The same construction works when $n = 0$ by using only the first term in (6.9).

6.3 An elementary proof of geometric acceleration

The bounds above already imply geometric convergence, without any saddle-point analysis. Since the central binomial coefficient is the largest of the $2N + 1$ coefficients whose sum is 4^N ,

$$\binom{2N}{N} \geq \frac{4^N}{2N+1}.$$

The $j = N$ summand in $T_{N,N+k}$ gives

$$T_{N,N+k} \geq \binom{2N}{N}^2 \geq \frac{16^N}{(2N+1)^2}.$$

Apply (6.2) and monotonicity to obtain, for $N \geq 1$,

$$0 < \zeta(3) - \Phi_{N,N+k} < \frac{(2N+1)^4}{2(N+k)^2 256^N}. \quad (6.10)$$

Here the tail has been taken in the larger coordinate $N + k$; the same estimate applied to the horizontal branch gives the weaker $(2N + 1)^4 / (2N^2 256^N)$, which (6.10) implies. Likewise, both $C_{N,N+k}$ and $C_{N+k,N}$ are at least $\binom{2N}{N}$. Using (6.8) for the first branch and the vertical bound (6.6) for the second gives

$$|\zeta(2) - \Psi_{N,N+k}| \leq \min \left\{ \frac{2(2N+1)^2}{(N+1)^2 16^N}, \frac{(2N+1)^2}{(N+k) 16^N} \right\}, \quad (6.11)$$

$$|\zeta(2) - \Psi_{N+k,N}| \leq \min \left\{ \frac{2(2N+1)^2}{(N+1)^2 16^N}, \frac{(2N+1)^2}{N 16^N} \right\}. \quad (6.12)$$

Neither branch dominates the other: the first decays like N^{-2} and so wins for large N , while the second is sharper in k for the upper diagonal at small N . These deliberately simple bounds are uniform over all offsets $k \geq 0$. The sharp exponential rates for fixed k are substantially faster.

6.4 Numerical certificates, not merely decimal agreement

table 1 reports rational bounds (6.4) and (6.9) computed by exact integer arithmetic. The displayed scientific notation has been rounded *upward*; the exact fractions are included in `data/certified_intervals.csv`. The full CSV also records the rational partial sums, the error signs, and decimal displays.

Table 1: Rigorous absolute-error upper bounds after N diagonal terms. Each displayed value is rounded upward to three significant digits.

Family	k	$N = 10$	$N = 20$	$N = 40$
$\zeta(3)$	0	$5.27 \cdot 10^{-30}$	$2.26 \cdot 10^{-60}$	$2.44 \cdot 10^{-121}$
	1	$1.79 \cdot 10^{-31}$	$7.15 \cdot 10^{-62}$	$7.45 \cdot 10^{-123}$
	3	$3.99 \cdot 10^{-34}$	$1.05 \cdot 10^{-64}$	$8.48 \cdot 10^{-126}$
$\zeta(2)$, upper	0	$4.57 \cdot 10^{-21}$	$5.88 \cdot 10^{-42}$	$9.46 \cdot 10^{-84}$
	1	$7.18 \cdot 10^{-22}$	$8.91 \cdot 10^{-43}$	$1.41 \cdot 10^{-84}$
	3	$2.69 \cdot 10^{-23}$	$2.59 \cdot 10^{-44}$	$3.53 \cdot 10^{-86}$
$\zeta(2)$, lower	0	$4.57 \cdot 10^{-21}$	$5.88 \cdot 10^{-42}$	$9.46 \cdot 10^{-84}$
	1	$2.80 \cdot 10^{-22}$	$3.44 \cdot 10^{-43}$	$5.40 \cdot 10^{-85}$
	3	$1.64 \cdot 10^{-24}$	$1.50 \cdot 10^{-45}$	$2.00 \cdot 10^{-87}$

For instance, the unshifted $\zeta(3)$ series after 40 terms has absolute error less than $2.44 \cdot 10^{-121}$. This is an absolute-error statement; claiming that a certain decimal string is correctly rounded would additionally require testing the certified interval against the relevant rounding boundary. The archive preserves exact fractions so that such tests can be made without relying on a floating-point zeta implementation. The next subsection carries out exactly that test.

6.5 From an interval to certified decimal digits

Let $[L, R]$ be either of the rational enclosures above and let d be a positive integer. The exact integer condition

$$\lfloor 10^d L \rfloor = \lfloor 10^d R \rfloor \quad (6.13)$$

certifies the first d decimal digits after the point of *every* number in $[L, R]$, and in particular of the constant enclosed. Both sides are computed by integer division, with no floating-point rounding

anywhere. This is the step deferred above: it turns an absolute-error bound into a statement about a decimal string.

The distinction matters, because a narrow interval need not certify many digits. An interval may have very small width and still straddle a decimal boundary, in which case (6.13) fails at a d well below $-\log_{10}(R - L)$. The accompanying program therefore tests (6.13) directly rather than reporting the width.

table 2 gives the resulting counts for $N = 60$ summands after the boundary harmonic sum. They are the actual common-prefix lengths of the generated intervals, not predictions from the leading asymptotics of section 8.

Table 2: Certified decimal places at $N = 60$, from (6.13) applied to the rational enclosures.

Offset k	$\zeta(3)$	$\zeta(2)$, upper	$\zeta(2)$, lower
0	181	124	124
1	182	125	125
3	186	127	128
10	194	131	136

These counts are consistent with the absolute-error table: $40 \cdot 3.0622 = 122.5$ and $40 \cdot 2.0899 = 83.6$ match the exponents reported above, and $60 \cdot 3.0622 = 183.7$ and $60 \cdot 2.0899 = 125.4$ match the $N = 60$ counts after the loss caused by boundary crossings. The exact interval endpoints are stored in `data/certified_digits.csv` and `data/decimal_certificates.json`.

7 Recovering the classical Apéry recurrences

The square identities also explain why these arrays contain the familiar Apéry denominators. This provides an independent consistency check and connects the proof with the usual one-dimensional formulation.

7.1 Denominator recurrences

On the diagonal of theorem 3.1,

$$T_{n,n-1} = T_{n-1,n} = \frac{A_n + A_{n-1}}{6} \quad (n \geq 1). \quad (7.1)$$

At the square with upper-right corner $(n + 1, n)$, (3.1) gives

$$(n + 1)^3 T_{n+1,n} + n^3 T_{n,n-1} = (2n + 1)(3n^2 + 3n + 1)A_n.$$

Substitute (7.1). After multiplication by 6 and collection of terms, the result is

$$(n + 1)^3 A_{n+1} = (34n^3 + 51n^2 + 27n + 5)A_n - n^3 A_{n-1}, \quad n \geq 1. \quad (7.2)$$

The initial values are $A_0 = 1$ and $A_1 = 5$.

Similarly, the two diagonal identities for C give

$$C_{n,n-1} = \frac{2b_n - b_{n-1}}{5}, \quad C_{n-1,n} = \frac{b_n + 2b_{n-1}}{5}. \quad (7.3)$$

Use (3.10) at $(m, n) = (n + 1, n)$:

$$(n + 1)^2 C_{n+1,n} + 2n^2 C_{n,n-1} = (n^2 + (2n + 1)^2) b_n.$$

Substitution from (7.3) gives

$$(n + 1)^2 b_{n+1} = (11n^2 + 11n + 3) b_n + n^2 b_{n-1}, \quad n \geq 1, \quad (7.4)$$

with $b_0 = 1$ and $b_1 = 3$. These are the classical recurrences recorded for A005259 and A005258 [3, 4].

7.2 The companion numerator solutions

Define rational arrays

$$P_{m,n} = T_{m,n} \Phi_{m,n}, \quad R_{m,n} = C_{m,n} \Psi_{m,n}. \quad (7.5)$$

They satisfy exactly the same local linear identities as T and C , respectively. For example, abbreviate the four Φ values by $\Phi_a, \Phi_b, \Phi_c, \Phi_d$. Then

$$\begin{aligned} m^3 d \Phi_d + n^3 a \Phi_a - s c \Phi_c &= m^3 d (\Phi_d - \Phi_c) + n^3 a (\Phi_a - \Phi_c) \\ &= m^3 d h_3(m, n) - n^3 a v_3(m - 1, n) \\ &= \frac{1}{c} - \frac{1}{c} = 0. \end{aligned}$$

The other identity follows by symmetry. For R , the corresponding cancellations are

$$\begin{aligned} 2n^2 d \Psi_d - m^2 a \Psi_a - q b \Psi_b &= 2n^2 d v_2(m, n) + m^2 a h_2(m, n - 1) = 0, \\ m^2 d \Psi_d + 2n^2 a \Psi_a - q c \Psi_c &= m^2 d h_2(m, n) - 2n^2 a v_2(m - 1, n) = 0. \end{aligned}$$

Thus the same elimination used for the denominators applies to their numerator companions.

Set $p_n = P_{n,n}$ and $r_n = R_{n,n}$. Then p_n satisfies (7.2), with $p_0 = 0, p_1 = 6$, and r_n satisfies (7.4), with $r_0 = 0, r_1 = 5$. In particular,

$$\frac{p_n}{A_n} = \Phi_{n,n} \longrightarrow \zeta(3), \quad \frac{r_n}{b_n} = \Psi_{n,n} \longrightarrow \zeta(2).$$

The adjacent determinant identities (often called Casoratians) are immediate from the diagonal increments:

$$p_n A_{n-1} - p_{n-1} A_n = \frac{6}{n^3}, \quad (7.6)$$

$$r_n b_{n-1} - r_{n-1} b_n = \frac{5(-1)^{n+1}}{n^2}. \quad (7.7)$$

The normalization here allows rational numerator sequences; no claim that all p_n or r_n are integers is intended. In the standard irrationality arguments, additional denominator-control statements are important. They are not needed to prove any of the series identities in this note.

7.3 A three-term recurrence on every shifted diagonal

The recurrences above are statements about the main diagonal. The same corner identities give a recurrence along each shifted diagonal as well, which is what one wants for computation: it avoids generating a two-dimensional array. The route is a row recurrence followed by one elimination.

Proposition 7.1 (Row recurrence). *For every $m \geq 1$ and every $n \geq 0$,*

$$(m+1)^3 T_{m+1,n} = (2m+1)(m^2 + m + 2n^2 + 2n + 1) T_{m,n} - m^3 T_{m-1,n}. \quad (7.8)$$

Proof. Let $n \geq 1$. Applying (3.1) to the square with upper-right corner $(m+1, n)$ gives

$$s(m+1, n) T_{m,n} = (m+1)^3 T_{m+1,n} + n^3 T_{m,n-1},$$

where $s(m, n) = (m+n)(m^2 + mn + n^2)$. The first identity in (3.8), at (m, n) , gives $n^3 T_{m,n-1} = m^3 T_{m-1,n} - \bar{s}(m, n) T_{m,n}$. Eliminating $T_{m,n-1}$ leaves (7.8), because

$$s(m+1, n) + \bar{s}(m, n) = (2m+1)(m^2 + m + 2n^2 + 2n + 1).$$

For $n = 0$ every entry is 1 and (7.8) reduces to the numerical identity $(m+1)^3 = (2m+1)(m^2 + m + 1) - m^3$. $\square$

Fix $k \geq 0$ and set, for $n \geq 0$,

$$D_n = T_{n,n+k}, \quad \sigma_n = s(n, n+k) = (2n+k)(3n^2 + 3nk + k^2), \quad w_n = n^3(n+k)^3, \quad (7.9)$$

$$G_n = (2n+1)(n^2 + n + 2(n+k)^2 + 2(n+k) + 1),$$

$$K_n = \sigma_{n+1}(\sigma_n G_n - n^6) - \sigma_n(n+1)^6. \quad (7.10)$$

Only $n \geq 1$ occurs in σ_n and w_n below, so both are positive there.

Theorem 7.2 (Shifted-diagonal recurrence). *For every $k \geq 0$ and every $n \geq 1$,*

$$w_{n+1} \sigma_n D_{n+1} = K_n D_n - \sigma_{n+1} w_n D_{n-1}, \quad (7.11)$$

with initial values

$$D_0 = 1, \quad D_1 = 2k^2 + 6k + 5. \quad (7.12)$$

The rational sequence $\widehat{D}_n = D_n S_{n,k}^{(3)}$ satisfies the same recurrence (7.11), with

$$\widehat{D}_0 = H_k^{(3)}, \quad \widehat{D}_1 = D_1 H_k^{(3)} + \frac{\sigma_1}{(k+1)^3}, \quad (7.13)$$

and the two solutions are linked by the discrete Wronskian

$$\widehat{D}_n D_{n-1} - \widehat{D}_{n-1} D_n = \frac{\sigma_n}{w_n} \quad (n \geq 1). \quad (7.14)$$

Proof. Write $m = n + k$. The identity (3.1) at (n, m) gives

$$T_{n-1,m} = \frac{n^3 D_n + m^3 D_{n-1}}{\sigma_n}.$$

Now apply (7.8) with dimension index n and radius index m , that is,

$$(n+1)^3 T_{n+1,m} = G_n T_{n,m} - n^3 T_{n-1,m}.$$

Substituting the previous display for $T_{n-1,m}$ and multiplying by σ_n yields

$$(n+1)^3 \sigma_n T_{n+1,m} = (\sigma_n G_n - n^6) D_n - w_n D_{n-1}.$$

Next, (3.2) at $(n+1, m+1)$ reads

$$\sigma_{n+1} T_{n+1,m} = (m+1)^3 D_{n+1} + (n+1)^3 D_n.$$

Eliminating $T_{n+1,m}$ between the two displays, and using $(n+1)^3(m+1)^3 = w_{n+1}$, gives (7.11). The initial values are $T_{0,k} = 1$ and, by symmetry and (2.5), $T_{1,k+1} = T_{k+1,1} = 1 + 2(k+1)(k+2) = 2k^2 + 6k + 5$.

For the companion, (4.5) gives $S_{n,k}^{(3)} - S_{n-1,k}^{(3)} = \sigma_n / (w_n D_{n-1} D_n)$, and multiplying by $D_n D_{n-1}$ is exactly (7.14). Divide (7.11) by $w_{n+1} \sigma_n$ and write it as $D_{n+1} = \alpha_n D_n - \beta_n D_{n-1}$ with $\beta_n = \sigma_{n+1} w_n / (\sigma_n w_{n+1})$. If W_n denotes the left side of (7.14), then $W_{n+1} = \beta_n W_n$, and

$$D_n (\widehat{D}_{n+1} - \alpha_n \widehat{D}_n + \beta_n \widehat{D}_{n-1}) = W_{n+1} - \beta_n W_n = 0.$$

Since $D_n > 0$, the companion satisfies the same recurrence. Its initial values follow from $S_{0,k}^{(3)} = H_k^{(3)}$ and the first diagonal increment. $\square$

At $k = 0$ we have $\sigma_n = 6n^3$, $w_n = n^6$ and $G_n = (2n+1)(3n^2 + 3n + 1)$, and dividing (7.11) by $6n^3(n+1)^3$ returns (7.2) exactly, since

$$\frac{K_n}{6n^3(n+1)^3} = 6(2n+1)(3n^2 + 3n + 1) - n^3 - (n+1)^3 = 34n^3 + 51n^2 + 27n + 5.$$

The Wronskian (7.14) likewise becomes $\sigma_n / w_n = 6/n^3$, which is (7.6), and $\widehat{D}_n = p_n$. Thus theorem 7.2 is the general case and the classical cubic recurrence is its $k = 0$ specialization.

Two limitations should be stated plainly. First, no analogue of theorem 7.2 for the quadratic array is proved here: the $\zeta(2)$ recurrence (7.4) is established only on the main diagonal, and an all-offset recurrence for $C_{n,n+k}$ is not supplied. Second, there is an arithmetic distinction between the two solutions. Every D_n is an integer, but $\widehat{D}_n$ need not be. Consequently the approximation estimates of section 6 are not, without an additional denominator argument, a self-contained new proof of irrationality; fast approximation and irrationality are not conflated anywhere in this note.

For computation along a fixed offset, (7.11) reaches index N in $O(N)$ recurrence steps rather than by generating a two-dimensional array. This counts arithmetic operations, not bit operations on integers whose length grows linearly in N .

8 Sharp growth and error asymptotics

The previous sections already prove the conjectures and give rigorous rational error bounds. We now determine the exact fixed-offset asymptotics, both to quantify the acceleration and to distinguish the effect of an offset from the effect of taking more diagonal terms. Every statement in this section has k fixed, with N or n tending to infinity. No estimate uniform in a growing offset is asserted anywhere.

Put

$$\tau = 1 + \sqrt{2}, \quad q_3 = \tau^2 = 3 + 2\sqrt{2}, \quad \lambda_3 = \tau^4 = 17 + 12\sqrt{2}, \quad (8.1)$$

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad \lambda_2 = \varphi^5 = \frac{11 + 5\sqrt{5}}{2}, \quad (8.2)$$

$$c_3 = \frac{q_3}{2^{9/4}\pi^{3/2}}, \quad c_2 = \frac{\varphi^{5/2}}{2\pi 5^{1/4}}. \quad (8.3)$$

Two elementary relations between these surds will simplify the error constants below:

$$\lambda_3^2 - 1 = 24\sqrt{2} \lambda_3, \quad \lambda_2^2 + 1 = 5\sqrt{5} \lambda_2. \quad (8.4)$$

Theorem 8.1 (Fixed-offset array growth). *For every fixed integer $k \geq 0$, as $N \rightarrow \infty$,*

$$T_{N,N+k} \sim c_3 q_3^k \lambda_3^N N^{-3/2}, \quad (8.5)$$

$$C_{N,N+k} \sim c_2 \varphi^{2k} \lambda_2^N N^{-1}, \quad (8.6)$$

$$C_{N+k,N} \sim c_2 \varphi^{3k} \lambda_2^N N^{-1}. \quad (8.7)$$

8.1 A discrete Laplace estimate

We use the following elementary form of the saddle-point principle. Suppose f has a unique interior maximum at $t_0 \in (0, 1)$, $f''(t_0) < 0$, and the summands near $j/N = t_0$ have the uniform form

$$N^{-a} g(j/N) e^{Nf(j/N)} (1 + o(1)), \quad g(t_0) > 0.$$

If summands outside every fixed neighborhood of t_0 are bounded by a polynomial in N times $e^{N(f(t_0)-\delta)}$ for some $\delta > 0$, then their sum is asymptotic to

$$g(t_0) \sqrt{\frac{2\pi}{|f''(t_0)|}} N^{1/2-a} e^{Nf(t_0)}. \quad (8.8)$$

To see this, expand $f(t) = f(t_0) + \frac{1}{2}f''(t_0)(t - t_0)^2 + O(|t - t_0|^3)$ and use $x = (j - Nt_0)/\sqrt{N}$. The local sum becomes a Riemann sum for the Gaussian integral with mesh $N^{-1/2}$; the complement is negligible by the strict maximum and a Gaussian majorant. First restrict to a fixed bounded x interval, then let that interval expand. This also justifies multiplying the summands by a fixed continuous function of j/N : the leading constant is multiplied by its value at t_0 .

8.2 The T saddle

For $j/N = t$ in a compact subinterval of $(0, 1)$, Stirling's formula gives

$$L(N, j) \sim \frac{1}{2\pi N t} \sqrt{\frac{1+t}{1-t}} e^{N\ell(t)}, \quad (8.9)$$

where

$$\ell(t) = (1+t)\log(1+t) - (1-t)\log(1-t) - 2t\log t.$$

The exponent for $T_{N,N}$ is $f_3(t) = 2\ell(t)$, and its amplitude is $N^{-2}g_3(t)$ with

$$g_3(t) = \frac{1+t}{4\pi^2 t^2 (1-t)}.$$

Since

$$\ell'(t) = \log \frac{1-t^2}{t^2},$$

the unique maximum is at $t_3 = 1/\sqrt{2}$. Direct substitution gives

$$e^{f_3(t_3)} = \lambda_3, \quad f_3''(t_3) = -8\sqrt{2}, \quad g_3(t_3) = \frac{q_3}{2\pi^2}.$$

Equation (8.8) now gives $c_3 \lambda_3^N N^{-3/2}$.

For a fixed offset, the exact ratio is

$$\frac{L(N+k, j)}{L(N, j)} = \prod_{r=1}^k \frac{N+j+r}{N-j+r}. \quad (8.10)$$

Near t_3 , this tends uniformly to $((1+t)/(1-t))^k$, whose value at t_3 is q_3^k . This proves (8.5).

For completeness, the endpoint estimates required by (8.8) follow directly from factorial bounds for the binomial summands. Their logarithms equal $N f_3(j/N) + O(\log N)$ uniformly when the entropy terms are extended continuously to 0 and 1. For fixed k , the ratio (8.10) is bounded by a polynomial in N on the whole summation range. Since f_3 has a strict interior maximum, the endpoints and every region separated from t_3 are exponentially smaller than the saddle contribution.

8.3 The C saddle and its asymmetry

Multiplying (8.9) by Stirling's approximation for $\binom{N}{j}$ gives exponent

$$f_2(t) = (1+t) \log(1+t) - 2(1-t) \log(1-t) - 3t \log t$$

and amplitude $N^{-3/2} g_2(t)$, where

$$g_2(t) = \frac{\sqrt{1+t}}{(2\pi)^{3/2} t^{3/2} (1-t)}.$$

The stationary equation is

$$f_2'(t) = \log \frac{(1+t)(1-t)^2}{t^3} = 0, \quad 1-t-t^2 = 0.$$

Thus the unique maximum occurs at $t_2 = 1/\varphi$. At that point,

$$e^{f_2(t_2)} = \lambda_2, \quad f_2''(t_2) = -\sqrt{5} \varphi^3,$$

and (8.8) gives $c_2 \lambda_2^N N^{-1}$.

For the upper diagonal, the offset multiplies the summand by

$$\frac{\binom{N+k}{j}}{\binom{N}{j}} = \prod_{r=1}^k \frac{N+r}{N-j+r} \longrightarrow (1-t)^{-k},$$

which equals φ^{2k} at t_2 . For the lower diagonal the multiplier is (8.10), with limiting value $((1+t_2)/(1-t_2))^k = \varphi^{3k}$. The same endpoint argument as for T applies. This proves [theorem 8.1](#), including the asymmetric constants in its last two formulas.

8.4 Exact leading constants for the errors

Let $S_{N,k}^{(3)} = \Phi_{N,N+k}$, and retain the notation $S_{N,k}^{(2,+)} = \Psi_{N,N+k}$ and $S_{N,k}^{(2,-)} = \Psi_{N+k,N}$.

Theorem 8.2 (Sharp fixed-offset errors). *For each fixed $k \geq 0$,*

$$\zeta(3) - S_{N,k}^{(3)} \sim 4\pi^3 \lambda_3^{-(2N+k+1)}, \quad (8.11)$$

$$\zeta(2) - S_{N,k}^{(2,+)} \sim (-1)^N 4\pi^2 \varphi^{-4k} \lambda_2^{-(2N+1)}, \quad (8.12)$$

$$\zeta(2) - S_{N,k}^{(2,-)} \sim (-1)^{N+k} 4\pi^2 \varphi^{-6k} \lambda_2^{-(2N+1)}. \quad (8.13)$$

In the signed formulas, $\sim$ means that the ratio to the displayed signed expression tends to 1.

Proof. Let $a_{n,k}$ be the positive summand in (2.6). By theorem 8.1,

$$a_{n,k} \sim \frac{6\lambda_3}{c_3^2 q_3^{2k}} \lambda_3^{-2n} = \frac{96\sqrt{2}\pi^3}{\lambda_3^k} \lambda_3^{-2n}.$$

The powers n^{-3} in the rational prefactor and in the product of array asymptotics cancel. Summing the resulting positive geometric tail divides by $\lambda_3^2 - 1$ and gives

$$\zeta(3) - S_{N,k}^{(3)} \sim \frac{96\sqrt{2}\pi^3}{\lambda_3^k(\lambda_3^2 - 1)} \lambda_3^{-2N} = 4\pi^3 \lambda_3^{-(2N+k+1)},$$

the last step by the first relation in (8.4).

For the upper $\zeta(2)$ series, the absolute value of the n th summand is asymptotic to

$$\frac{5\lambda_2}{c_2^2 \varphi^{4k}} \lambda_2^{-2n} = \frac{20\pi^2 \sqrt{5}}{\varphi^{4k}} \lambda_2^{-2n}.$$

For the lower series, replace φ^{4k} by φ^{6k} and include the additional sign $(-1)^k$. An alternating geometric tail has denominator $1 + \lambda_2^{-2}$, so the constants are $20\pi^2 \sqrt{5}/(\varphi^{4k}(\lambda_2^2 + 1))$ and $20\pi^2 \sqrt{5}/(\varphi^{6k}(\lambda_2^2 + 1))$; the second relation in (8.4) reduces these to the forms displayed in (8.12) and (8.13).

To justify these tail operations directly, if $a_n/(K\rho^n) \rightarrow 1$ with $0 < \rho < 1$, then $\sup_{n>N} |a_n/(K\rho^n) - 1| \rightarrow 0$. The absolute error incurred by replacing the entire tail with the corresponding geometric tail is therefore $o(\rho^N)$, in both the positive and alternating cases. No exchange of an uncontrolled asymptotic expansion with a sum is used. $\square$

The asymptotic decimal gain per extra diagonal term is

$$2 \log_{10} \lambda_3 = 3.0622054827 \dots \quad \text{for } \zeta(3), \quad 2 \log_{10} \lambda_2 = 2.0898764025 \dots \quad \text{for } \zeta(2).$$

A fixed offset changes the leading constant, not these exponential rates. Increasing k also increases the size and cost of the array entries, so the prefactor improvement alone does not establish an optimal computational strategy.

8.5 Why a fixed row is slower than a diagonal

For comparison, fix m and let $n \rightarrow \infty$. The leading binomial terms give

$$T_{m,n} \sim \alpha_m n^{2m}, \quad \alpha_m = \frac{\binom{2m}{m}}{(m!)^2}; \quad C_{m,n} \sim \beta_m n^m, \quad \beta_m = \frac{\binom{2m}{m}}{m!}.$$

Using the exact vertical tails and the elementary asymptotic $\sum_{j>n} j^{-s} \sim n^{1-s}/(s-1)$ for $s > 1$, we obtain

$$\zeta(3) - \Phi_{m,n} \sim \frac{n^{-(4m+2)}}{\alpha_m^2(4m+2)}, \quad (8.14)$$

$$\zeta(2) - \Psi_{m,n} \sim (-1)^m \frac{n^{-(2m+1)}}{\beta_m^2(2m+1)}. \quad (8.15)$$

Thus a fixed row gives increasing algebraic acceleration as the row index grows, whereas a fixed-offset diagonal gives geometric acceleration. Both behaviors are consequences of the same potential.

9 Reproducible verification and computational use

The proof is finite algebra followed by explicit convergence estimates. The accompanying computations are a separate audit of indexing, normalization, and signs. In particular, the proof of an infinite identity is not inferred from decimal agreement.

9.1 What was checked

The program `verify.py` uses only Python's standard library, with arbitrary-precision integers and rational fractions. It is the union regression suite of this archive: it runs, in one pass and under one total, every group of checks that either source package contributed. Its default run checks the following exact equalities:

Test group	Equalities
Four-corner contiguity identities, $1 \leq m, n \leq 48$	9,216
Closed squares and diagonal increments, $1 \leq m, n \leq 40$	6,400
Termwise telescoper identities, $1 \leq m, n \leq 16$	7,008
Potentials, symmetry, and independent array formulas	8,405
Finite diagonal formulas, offsets $0 \leq k \leq 12$	1,599
Random monotone lattice paths, 300 per potential	600
Row recurrence (7.8), $1 \leq m \leq 40, 0 \leq n \leq 20$	840
Shifted-diagonal recurrence, companion, and Wronskian	670
Main-diagonal denominator and numerator recurrences	156
Published diagonal prefixes	2
Certified decimal prefixes and interval intersections	14
Total	34,910

The potential and independent array tests use $0 \leq m, n \leq 40$. Finite diagonal formulas are checked for every $0 \leq N \leq 40$ and $0 \leq k \leq 12$, including all three orientations/families and the empty-sum cases. The random-path tests draw 300 monotone words with a fixed seed, 20260920, and compare the accumulated edge weights of each word with the potential at its endpoint; they exercise path independence itself rather than only the square identities. The termwise tests check the telescoping certificates summand by summand, including the terminating endpoint $j = \min(m, n)$ at which a factor vanishes.

For additional independence, C is compared with the coefficient formula proved in [section B](#), and T is compared with the alternative OEIS sum

$$\sum_{j=0}^{\min(m,n)} \binom{m}{j}^2 \binom{m+n-j}{m}^2$$

[1]. These comparisons are finite implementation tests; the proof of the conjectures uses (2.2) and (2.3) directly.

Finite tests cannot establish the infinitely quantified theorems. Their purpose is to expose transcription, indexing, sign, or implementation errors independently of the written derivations. The exact bounds of [section 6](#) prove the rational enclosures; the regression tests themselves do not.

9.2 Two independent implementations

This archive keeps a second, deliberately independent implementation, `verify_independent.py`, written in the transposed naming convention of the other source package: it calls the A108625 array U and writes its index pair with the dimension first as (n, m) . Its own run reports 25,840 checks. Where both programs test the same statement they do so from separately written formulas, which is the point of retaining both; the headline count above is the union suite's.

The two symbolic suites are kept for the same reason. `symbolic_certificates.py` uses SymPy to reduce the three rational certificates (3.4), (3.11), (3.12), and the two recurrence-elimination coefficients, to zero; all five pass. `symbolic_certificates_independent.py` reduces seven further rational-function identities to zero: the antisymmetric relation behind (3.5), the symmetric relation behind (3.6), the conjugate products of (3.7) and (3.13), the two residuals for (3.9) and (3.10), and the row-recurrence coefficient of [theorem 7.1](#). The supplied records identify the tested SymPy version. These are symbolic identities, not evaluations at sample numerical points.

9.3 How to reproduce the artifacts

From the archive directory, run

```
python verify.py --max-index 40 --offsets 12 --output-dir data
python make_error_table.py
python verify_independent.py
python symbolic_certificates.py
python symbolic_certificates_independent.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The two symbolic commands require SymPy; they are not required for the first three commands or for compiling the supplied article. The optional `numerical_checks.py` compares the exact partial

sums with high-precision zeta values, and `asymptotic_diagnostics.py` compares the array and error asymptotics of [section 8](#) with 420-digit mpmath evaluations. Both require mpmath, and both are labeled diagnostics: their output is not used as an error certificate.

The CSV interval data store numerators and denominators rather than only decimal approximations. For $\zeta(3)$, use the endpoints S and $S + B$. For $\zeta(2)$, the recorded sign $\epsilon \in \{1, -1\}$ gives endpoints S and $S + \epsilon B$, ordered increasingly. The certified decimal prefixes of [table 2](#) are regenerated into `data/certified_digits.csv` by `verify.py` and, with the independent program's own bound, into `data/decimal_certificates.json`. The latter uses only the largest-coordinate branch (6.3) and therefore certifies one place fewer for $\zeta(3)$ at $k = 0$ and $k = 3$; both counts are correct for their respective bounds. The article, source notes, test programs, and generated data are all included in the archive.

9.4 A practical evaluation route

A direct array entry is a finite sum of $\min(m, n) + 1$ positive integers. The summand ratios in [section 3](#) allow successive summands to be generated without repeatedly evaluating factorials. For a fixed diagonal and N terms, the direct implementation uses $O(N^2)$ summand-generation steps when k is fixed; this counts arithmetic operations, not bit complexity. The integers grow exponentially in value and hence linearly in bit length along a fixed-offset diagonal.

For the main diagonals, (7.2) and (7.4) reduce denominator generation to $O(N)$ recurrence steps. The same recurrences for p_n and r_n produce the associated rational approximants. Whichever method is used, one can stop when the rational bound is below a prescribed absolute tolerance. The positive binomial formulas are particularly convenient for independent verification and avoid relying on cancellation in a recurrence.

10 Conclusion and scope

The conjectural family in A143007 is true for every nonnegative offset. Both neighboring-diagonal conjectures in A108625 are also true, and belong to two full offset families. The proof does not assume any of the conjectural formulas, a known diagonal limit, or Apéry's irrationality theorem. Instead it identifies the limit from the ordinary convergent series for $\zeta(3)$ and $\zeta(2)$ on the coordinate axes.

The structural reason is a closed rational system of edge increments. A finite telescoping certificate yields each square identity; the square identities yield path independence; a two-edge diagonal step cancels its intermediate array entry; and the boundary limits identify the sum. This chain of implications explains the cubic and quadratic numerators in the OEIS formulas, as well as the differing constants and signs in the two $\zeta(2)$ orientations.

Several further consequences come out of the same argument: exact horizontal and vertical remainders, convergence along every unbounded monotone lattice path, signed rational bounds, certified decimal prefixes, a three-term recurrence on each shifted cubic diagonal, and sharp fixed-offset leading constants. All limiting steps are supplied after, rather than assumed from, the local algebra.

The conclusions concern the stated acceleration formulas; no new irrationality theorem or resolution of unrelated Apéry-polynomial conjectures is claimed. In particular, this work does not settle the irreducibility conjectures in the classical Apéry entries or the other congruence conjectures listed in the two array entries; it does not claim a new proof of irrationality from fast

convergence alone; it is not a proof-assistant formalization, and the accompanying finite checks do not replace mathematical proof; and, because related conservative fields and their conjugate polynomials already occur in [5], no historical novelty or priority is claimed for the framework or for consequences that may already be implicit in that literature. The specific accomplishment is an explicit, self-contained derivation of the formulas that the consulted OEIS entries still label conjectural.

A A compact route through the proof

A reader interested only in the conjectural identities can omit [sections 7](#) and [8](#) entirely. The logically sufficient chain is

$$(3.3), \text{ the ratios before (3.11)} \implies \text{theorem 3.1, theorem 3.2} \implies (4.1),$$

that is, the two telescoping certificates give the four-corner identities, and the four-corner identities give square closure for the weights (4.2) and (5.1). [theorem 4.1](#) then constructs the potentials Φ and Ψ ; the elementary boundary estimates of [theorems 4.2](#) and [5.1](#) identify their limits; and the diagonal increments (4.4) and (5.6) telescope to [theorems 2.1](#) to [2.3](#). The sum certificates vanish at both endpoints, all corner denominators are positive, and equal coordinates are permitted throughout.

B Identification with the OEIS arrays and row generating functions

This appendix verifies the array identification algebraically and explains the Legendre-polynomial structure behind the crystal-ball interpretation. It is independent of the potential proof.

B.1 Two finite forms of the Legendre polynomial

Define the Legendre polynomial by Rodrigues' formula,

$$P_m(z) = \frac{1}{2^m m!} \frac{d^m}{dz^m} (z^2 - 1)^m.$$

Expansion at $z = 1$ gives

$$P_m(1 + 2u) = \sum_{j=0}^m L(m, j) u^j. \quad (\text{B.1})$$

Indeed, after substituting $z = 1 + 2u$, Rodrigues' formula becomes $m!^{-1} (d/du)^m [u^m (1 + u)^m]$, whose coefficient of u^j is $\binom{m}{j} \binom{m+j}{j}$.

Applying the Leibniz rule instead gives

$$P_m(z) = 2^{-m} \sum_{i=0}^m \binom{m}{i}^2 (z-1)^i (z+1)^{m-i}. \quad (\text{B.2})$$

With $W_m(x) = \sum_{i=0}^m \binom{m}{i}^2 x^i$, this implies

$$P_m\left(\frac{1+x}{1-x}\right) = \frac{W_m(x)}{(1-x)^m}. \quad (\text{B.3})$$

B.2 The row generating function of C

Using $\sum_{n \geq 0} \binom{n}{j} x^n = x^j / (1-x)^{j+1}$ in (2.3), we obtain, as a formal power series,

$$\begin{aligned} \sum_{n \geq 0} C_{m,n} x^n &= \frac{1}{1-x} \sum_{j=0}^m L(m, j) \left(\frac{x}{1-x}\right)^j \\ &= \frac{1}{1-x} P_m\left(\frac{1+x}{1-x}\right) = \frac{W_m(x)}{(1-x)^{m+1}}. \end{aligned} \quad (\text{B.4})$$

Extracting the coefficient of x^n gives

$$C_{m,n} = \sum_{i=0}^{\min(m,n)} \binom{m}{i}^2 \binom{m+n-i}{m}. \quad (\text{B.5})$$

This is exactly the defining finite sum and row generating function listed in A108625 [2]. In particular, the orientation of the two indices has been verified, rather than inferred from a short sequence prefix.

B.3 A finite square identity and the row generating function of T

We next prove the finite identity

$$P_m(z)^2 = \sum_{j=0}^m L(m, j) \binom{2j}{j} \left(\frac{z^2 - 1}{4} \right)^j. \quad (\text{B.6})$$

It is a terminating Legendre square identity. A short differential verification is included to keep the identification self-contained.

Put $M = m(m+1)$ and $Q(z) = P_m(z)^2$. The Legendre equation $(1-z^2)P_m'' - 2zP_m' + MP_m = 0$, obtained by differentiating Rodrigues' formula, yields the following equation by the product rule:

$$(z^2 - 1)^2 Q''' + 6z(z^2 - 1)Q'' + ((6 - 4M)z^2 + 4M - 2)Q' - 4MzQ = 0. \quad (\text{B.7})$$

Since $P_m(-z) = (-1)^m P_m(z)$, Q is a polynomial in z^2 . Set $t = (z^2 - 1)/4$ and write $Q(z) = y(t)$. Substitution in (B.7) and cancellation as a polynomial identity gives

$$t^2(4t + 1)y''' + (18t^2 + 3t)y'' + ((12 - 4M)t + 1)y' - 2My = 0. \quad (\text{B.8})$$

For $y(t) = \sum_{j \geq 0} q_j t^j$, coefficient comparison gives

$$(j+1)^3 q_{j+1} = 2(2j+1)(m-j)(m+j+1)q_j, \quad q_0 = P_m(1)^2 = 1.$$

The coefficients $q_j = L(m, j) \binom{2j}{j}$ satisfy exactly this recurrence and initial value, and vanish for $j > m$. This proves (B.6).

Now

$$\sum_{n \geq 0} L(n, j) x^n = \binom{2j}{j} \frac{x^j}{(1-x)^{2j+1}},$$

so (2.2) and (B.6) imply

$$\begin{aligned} \sum_{n \geq 0} T_{m,n} x^n &= \frac{1}{1-x} \sum_{j=0}^m L(m, j) \binom{2j}{j} \left(\frac{x}{(1-x)^2} \right)^j \\ &= \frac{1}{1-x} P_m \left(\frac{1+x}{1-x} \right)^2 = \frac{W_m(x)^2}{(1-x)^{2m+1}}. \end{aligned} \quad (\text{B.9})$$

These are the row generating functions recorded in A143007. The passage from $W_m(x)/(1-x)^m$ to its square is exactly what the sum norm on a product lattice requires for sphere counts; division by $1-x$ then converts sphere counts to ball counts [1, 9].

References

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